

24-Channel Cryogenic Breakout Box

Shielded Interface for Low-Noise
Cryogenic & Quantum Measurements

Shielded Signal Integrity.
Cryogenic Compatible.
Engineered for Quantum Research.



CryoConnect is a fully shielded 24-channel cryogenic breakout box engineered for low-temperature quantum transport and quantum computing experiments. It provides a controlled interface between cryostat wiring and room-temperature instrumentation, effectively extending the cryostat Faraday cage to the measurement boundary. This architecture preserves signal integrity, minimizes electromagnetic interference (EMI), and ensures reliable operation of sensitive quantum and low-noise devices.

KEY FEATURES

- **Signal Handling & Channel Architecture**
24 independent shielded channels designed for scalable low-noise cryogenic measurements
- **Integrated Filtering**
Per-channel low-pass filtering for RF noise suppression and improved signal stability
- **Shielding & EMI Control**
Fully enclosed Faraday cage architecture for minimized electromagnetic interference
- **Low-Noise Wiring Design**
Twisted-pair signal routing optimized for reduced crosstalk and high-fidelity measurements
- **Grounding & Protection**
Configurable grounding system with enhanced protection for ESD-sensitive devices

Notes

- Designed for low-noise cryogenic measurement environments
- Performance dependent on overall system grounding and shielding practices
- Recommended for use with properly shielded cryostats and instrumentation
- Custom channel configurations and filtering options available

APPLICATIONS

- Quantum transport measurements
- Quantum computing device characterization
- Low-noise electrical measurements in cryogenic systems
- Multi-terminal device testing
Sensitive signal routing and conditioning
- Experimental setups requiring EMI-controlled environments



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SPECIFICATIONS

CHANNEL & SIGNAL ARCHITECTURE

Number of Channels	24
Channel Type	Individually shielded
Signal Routing	Twisted-pair
Crosstalk Control	Low-noise optimized design

FILTERING & NOISE SUPPRESSION

Filter Type	Integrated low-pass (per channel)
Purpose	RF noise suppression
Noise Control	Reduces high-frequency interference

SHIELDING & GROUNDING

Shielding	Full enclosure (Faraday cage extension)
Grounding	Per-channel switchable ground
Common Bus	Selectable shared grounding
ESD Protection	Integrated via grounding architecture

INTERFACES & CONNECTIVITY

Connector Type	BNC (front panel)
Interface Type	Standard laboratory instrumentation
Access	Direct channel-wise connectivity

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